

A Multi-Sensor AI Fusion Framework for Real-Time Crop Quality Grading and Post-Harvest Decision Support

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Abstract

Post-harvest quality assessment is often fragmented across visual inspection, individual measurement devices, and laboratory-based testing. This creates practical limitations for small-scale farmers who need rapid information about crop quality, moisture condition, storage suitability, and dispatch decisions. This paper presents AgriScan AI Pro, a proposed low-cost, portable, edge-computing framework that combines computer vision with synchronized physical and environmental sensing for post-harvest crop assessment. The architecture uses a Raspberry Pi 5 as the edge-processing unit and an ESP32-S3 as the sensor controller. A Raspberry Pi Camera 3 captures uniformly illuminated crop images, while a load cell with HX711, a grain moisture sensor, and a DHT22 provide weight, moisture, temperature, and relative-humidity measurements. The framework fuses visual and sensor-derived features and applies a crop-specific rule-based decision layer to generate quality grades, moisture status, storage/transport guidance, and post-harvest recommendations. The proposed workflow includes dataset preparation, image pre-processing, model training and validation, sensor calibration, synchronized data acquisition, multimodal fusion, and reference-based evaluation. The current manuscript deliberately does not report fabricated accuracy, error, freshness, or processing-time values; these quantities must be obtained from the experimental validation described in the paper before journal submission. The framework is intended to provide a practical foundation for affordable AI-enabled post-harvest decision support.

Keywords: Artificial intelligence; sensor fusion; computer vision; crop quality grading; post-harvest management; edge computing; smart agriculture; decision support system.

1. Introduction

Agricultural profitability depends not only on production yield but also on the quality retained after harvesting. Before storage, transportation, processing, or sale, harvested produce may need to be assessed for moisture condition, physical quality, visible defects, maturity, freshness, and environmental exposure. Errors or delays in these assessments can contribute to quality deterioration and post-harvest losses.

Conventional assessment frequently combines visual inspection with separate measurement instruments. Visual inspection can be subjective, while laboratory methods may require additional time, equipment, and skilled personnel. Separate moisture and weight measurements provide useful numbers but do not necessarily translate those measurements into an actionable recommendation such as whether a batch should be dried, stored, transported, or sold.

Recent work demonstrates the value of computer vision, deep learning, sensor-based measurement, and IoT architectures for agricultural quality assessment. Multimodal systems are particularly relevant because visual information describes observable surface characteristics while physical and environmental sensors provide quantitative measurements that are not available from an RGB image alone [1–5].

The research problem addressed in this study is therefore the integration of these heterogeneous information sources into a compact, edge-based system that can acquire data from the same sample, combine the measurements, and present an interpretable post-harvest decision to the user. AgriScan AI Pro addresses this problem through a dual-controller architecture in which Raspberry Pi 5 performs image processing, AI inference, data fusion, and reporting, while ESP32-S3 handles real-time sensor acquisition.

The main contributions of this work are: (i) a low-cost portable architecture for multimodal crop-quality assessment; (ii) synchronized acquisition of image, weight, moisture, temperature, and humidity information; (iii) an edge-based fusion and decision-support workflow; (iv) a crop-specific rule layer that translates measurements and AI outputs into practical post-harvest recommendations; and (v) a reproducible validation protocol for comparing the integrated system with reference measurements and unimodal baselines.

1.1 Research Objectives

- To design and implement a portable post-harvest assessment platform integrating computer vision and physical/environmental sensors.
- To acquire visual and sensor measurements from the same crop sample through synchronized data collection.
- To develop a computer-vision pipeline for crop identification and observable defect/quality assessment.
- To fuse AI-derived visual information with weight, moisture, temperature, and humidity measurements at the edge.
- To generate interpretable quality grades and post-harvest recommendations using crop-specific rules.
- To evaluate the system using a labelled dataset, calibrated reference measurements, and comparisons with vision-only and sensor-only baselines.

1.2 Research Questions

- Can synchronized visual and sensor measurements provide a more informative crop-quality assessment than a single measurement modality?
- Can the proposed edge architecture perform the complete acquisition, inference, fusion, and reporting workflow without requiring continuous cloud processing?
- How accurately can the integrated system reproduce reference crop-quality and moisture assessments under controlled test conditions?

2. Related Work

Automated agricultural quality assessment has progressed from manual inspection toward computer vision, machine learning, sensor-based monitoring, and multimodal systems. A 2026 review of AI and machine learning for post-harvest fruit-quality assessment highlights the growing role of automated approaches while also identifying challenges associated with generalisation, data quality, and practical deployment [1].

Mothkur et al. proposed AgroQuali-FSL, a few-shot deep-learning framework for automated grading of fruits and vegetables, demonstrating the relevance of deep learning to quality classes and visible defects [2]. Pugazhendi et al. reported a multimodal, non-destructive approach for pomegranate quality assessment that combined visual and physical measurements. Their reported integrated accuracy was 86.57%, compared with 74.07% for vision-only and 29.63% for physical-measurement-only approaches [3]. This provides direct motivation for evaluating multimodal fusion rather than relying on a single modality.

Perales Gómez et al. presented FARMIT, an IoT architecture for continuous crop-quality assessment using machine learning and deep learning with multiple data sources [4]. The work illustrates the value of aggregating sensor and imaging information, although distributed architectures can increase implementation complexity. Brilliantina et al. developed a non-destructive durian maturity-detection system using multiple sensor modalities and supervised machine learning, further supporting the use of multimodal sensing for agricultural quality assessment [5].

The literature indicates that visual AI, physical sensing, environmental monitoring, and IoT can each contribute useful information. However, the present work focuses on a compact edge implementation that combines visual, physical, and environmental inputs and converts them into a single farmer-facing decision support output. The novelty claim of this paper is therefore the system-level integration and validation workflow rather than a claim of a new deep-learning architecture.

Ref.	Primary focus	Main technology	Limitation relevant to this work
[1]	Post-harvest fruit-quality assessment	AI/ML review	Identifies practical challenges in automated quality assessment
[2]	Fruit and vegetable quality grading	Few-shot deep learning + computer vision	Primarily image-based quality grading
[3]	Pomegranate quality assessment	Vision + physical measurements	Focused on a single crop and specific physical features
[4]	Continuous crop-quality assessment	IoT + ML/DL	More distributed architecture
[5]	Durian maturity detection	Multivariable sensor + ML	Focused on maturity assessment of a specific fruit

3. Materials and Methods

The proposed AgriScan AI Pro architecture follows five logical stages: (1) data acquisition, (2) local preprocessing and AI inference, (3) sensor validation and normalization, (4) multimodal fusion and decision generation, and (5) user-facing reporting. The architecture is designed to keep the compute-intensive image pipeline separate from continuous sensor polling.

3.1 System Architecture

Raspberry Pi 5 is used as the main edge-computing unit for image capture, AI inference, feature/data fusion, and report generation. ESP32-S3 operates as the dedicated sensor node and continuously acquires weight, moisture, temperature, and humidity values before transmitting them to the Raspberry Pi through a serial interface. The separation of responsibilities reduces interference between sensor polling and compute-intensive image processing.

3.2 Hardware Configuration

Component	Role
Raspberry Pi 5	Edge processing, AI inference, data fusion, decision logic and report generation
Raspberry Pi Camera 3	Crop-image acquisition under controlled lighting
ESP32-S3	Sensor acquisition and serial data transmission
5 kg Load Cell + HX711	Sample weight measurement
Grain moisture sensor	Moisture measurement for storage/drying decisions
DHT22	Ambient temperature and relative-humidity measurement
3.5-inch touchscreen	Display of assessment results and recommendations
Push button	Scan/start interaction
Buzzer and RGB LED	Status and user feedback
Acrylic/PVC enclosure + fixed tray	Compact housing and consistent sample positioning

3.3 Data Acquisition

When the user initiates a scan, the system starts two parallel streams. The camera captures a uniformly illuminated image of the sample, while ESP32-S3 reads the sensor values. A single scan record should contain a unique sample identifier, timestamp, crop class, image path, weight, moisture, temperature, humidity, reference label, and system prediction. Synchronizing these streams is important because the visual and physical measurements must correspond to the same sample instance.

3.4 Image Pre-processing and AI Pipeline

The image pipeline should include image quality checking, resizing to the model input dimensions, pixel normalization, and application-specific augmentation during training. The model should be trained using strictly separated training, validation, and test sets. Samples from the same physical batch should not be split across these sets because such leakage can artificially inflate test performance.

The computer-vision task is defined as crop classification and observable quality/defect assessment. Depending on the final labelled dataset, defect categories may include fungal symptoms, discoloration, insect damage, or other visible quality classes. Ripeness/freshness estimation should only be reported for crop classes for which reliable labels and a defensible reference procedure are available.

3.5 Sensor Processing and Calibration

Raw sensor values are converted to engineering units and checked for missing, out-of-range, or unstable measurements. The load-cell system requires calibration against known masses. The moisture sensor should be compared with a suitable reference moisture measurement under the intended crop conditions. DHT22 temperature and humidity readings should be compared with a reference instrument before experimental use.

For a measured quantity x_i and reference value r_i , absolute error and relative error can be calculated as:

$$\text{Absolute error} = |x_i - r_i|$$

$$\text{Relative error (\%)} = |x_i - r_i| / |r_i| \times 100$$

For repeated measurements, mean absolute error (MAE) and root mean square error (RMSE) should be reported where appropriate. Calibration data must be collected independently of the final test data.

3.6 Multimodal Data Fusion

The fusion layer receives visual predictions/features and normalized sensor measurements. A practical implementation can use a structured feature vector containing crop class probabilities, defect indicators, weight, moisture, temperature, and relative humidity. The final quality grade is produced only after all required inputs pass validity checks.

For a crop-specific quality score S , the implementation should define the feature weights and thresholds before the final test set is evaluated. A generic weighted formulation is:

$$S = \sum(w_j \times z_j), \text{ where } \sum w_j = 1$$

where z_j represents a normalized feature contribution and w_j is its predefined weight. If a rule-based system is used instead of a learned fusion model, every threshold and rule must be documented and fixed before the final evaluation.

3.7 Decision Support and Recommendations

The decision layer maps validated inputs to interpretable outputs. The intended output categories are quality grade (A/B/C), moisture status (ideal/too wet/too dry), storage/transport suitability, and post-harvest recommendations. Recommendations such as drying time, safe storage period, seed suitability, shelf-life estimation, and dispatch timing should be generated only where crop-specific thresholds or validated agronomic rules are available.

3.8 Experimental Design and Validation Protocol

The experimental study should use a predefined set of crop samples covering the quality classes that the system claims to recognize. Each sample should be measured using the AgriScan system and an appropriate reference method. The test protocol should include repeated measurements to assess repeatability and should cover realistic variation in moisture, visible quality, and sample weight.

- Dataset: record the number of crops, images, classes, defect categories, and samples per class.
- Data split: use training, validation, and held-out test sets with no sample/batch leakage.
- Vision evaluation: report accuracy, precision, recall, F1-score and a confusion matrix for classification.
- Sensor evaluation: report MAE, RMSE and relative error against reference instruments where applicable.
- Fusion evaluation: compare vision-only, sensor/rule-only, and multimodal fusion performance.

- System evaluation: record end-to-end inference/processing time over repeated scans.
- Repeatability: repeat measurements on the same sample or controlled replicate samples and report variability.
- Ablation: where feasible, remove one modality at a time to quantify its contribution to the integrated result.

3.9 Statistical Reporting

Results should be reported as mean $\pm$ standard deviation for repeated continuous measurements where appropriate. Confidence intervals and statistical tests should be selected according to the experimental design and sample size. The final manuscript should state the number of independent samples and repeated measurements used for each analysis.

4. Results and Discussion

The current source manuscript describes the integrated prototype architecture and the intended processing workflow, but it does not provide experimentally measured dataset size, model performance, sensor error, repeatability, or end-to-end processing time. Therefore, those quantities are intentionally not fabricated in this revised manuscript. The following subsections define the exact results that must be inserted from the completed experiments.

4.1 Prototype Operation

The intended scan sequence begins with user activation, followed by simultaneous image and sensor acquisition. The Raspberry Pi processes the image while the ESP32-S3 supplies synchronized physical and environmental measurements. The fusion layer then produces a consolidated assessment and the touchscreen displays the result. The architecture shown in the original manuscript provides a clear separation between data acquisition, processing, AI, output, and user-interface stages.

4.2 Dataset Summary — REQUIRED EXPERIMENTAL DATA

Item	Value to report
Number of crop classes	Not measured in the current study
Total images	Not measured in the current study
Training images	Not measured in the current study
Validation images	Not measured in the current study
Test images	Not measured in the current study
Defect/quality classes	To be defined from the final experimental dataset
Independent physical samples	Not measured in the current study

4.3 AI Model Performance — REQUIRED EXPERIMENTAL DATA

Model/task	Accuracy	Precision	Recall	F1-score
Crop classification	Not measured	Not measured	Not measured	Not measured
Quality/defect classification	Not measured	Not measured	Not measured	Not measured

A confusion matrix should be included as a figure. If different crops use different defect labels, report the class definitions and per-class metrics rather than combining incompatible labels.

4.4 Sensor Validation — REQUIRED EXPERIMENTAL DATA

Measurement	Reference method	MAE	RMSE	Relative error
Weight	Reference instrument to be used in validation	Not measured	Not measured	Not measured %
Moisture	Reference instrument to be used in validation	Not measured	Not measured	Not measured %
Temperature	Reference instrument to be used in validation	Not measured	Not measured	Not measured %
Relative humidity	Reference instrument to be used in validation	Not measured	Not measured	Not measured %

4.5 Multimodal Fusion Comparison — REQUIRED EXPERIMENTAL DATA

Approach	Accuracy	F1-score	Key observation
Vision only	Not measured	Not measured	Not evaluated quantitatively
Sensor/rule only	Not measured	Not measured	Not evaluated quantitatively
Vision + sensor fusion	Not measured	Not measured	Not evaluated quantitatively

4.6 Processing Time — REQUIRED EXPERIMENTAL DATA

Stage	Mean time (s)	Std. dev. (s)
Image capture	Not measured	Not measured
AI inference	Not measured	Not measured
Sensor acquisition	Not measured	Not measured
Fusion + decision	Not measured	Not measured
Total end-to-end scan	Not measured	Not measured

4.7 Discussion

The central technical question is whether combining visual and sensor information provides measurable benefit over unimodal assessment. The final discussion should therefore compare the three evaluation conditions, explain failure cases, and identify which inputs contribute most to the decision. The discussion should also distinguish between visible defects that can be detected from RGB images and latent properties such as moisture that require physical measurement.

The system should not claim laboratory-equivalent performance unless the experimental design directly supports such a conclusion. Likewise, shelf-life, seed viability, drying time, and economic benefit should be presented as decision-support estimates only when crop-specific reference data and validation procedures are available.

4.8 Prototype Demonstration and Current Evidence Status

The present study establishes and documents the AgriScan AI Pro prototype architecture, sensing workflow, edge-processing pipeline, multimodal fusion logic, and decision-support mechanism. The prototype is designed to acquire crop images together with weight, moisture, temperature, and humidity measurements and to present an integrated post-harvest assessment.

The current manuscript intentionally does not claim experimentally measured classification accuracy, sensor error, freshness percentage, shelf-life accuracy, or end-to-end processing time. These quantities require controlled experiments with labelled wheat samples and suitable reference instruments. Reporting invented numerical values would not constitute valid scientific evidence.

Accordingly, the contribution of the present manuscript is positioned as a system design and prototype methodology. The validation protocol in this paper defines how quantitative performance should be established in subsequent controlled experiments, including held-out AI testing, reference-sensor comparison, multimodal ablation, repeatability testing, and end-to-end latency measurement.

5. Limitations and Threats to Validity

- The present source manuscript does not yet contain the experimental dataset or quantitative validation results.
- RGB computer vision is limited to observable characteristics and may miss internal defects or properties not expressed on the surface.
- Moisture-sensor performance can vary with crop type, sample packing, temperature, and calibration conditions.
- Rule-based recommendations depend on the correctness and crop-specific validity of the thresholds used.
- A model trained on limited crops or controlled lighting may not generalize to different cultivars, illumination conditions, camera distances, or field environments.
- Performance claims should therefore be restricted to the crops, conditions, hardware, and dataset represented in the final experiments.

6. Conclusion

This revised manuscript presents AgriScan AI Pro as a low-cost, portable, edge-based framework for multimodal post-harvest crop assessment. The system integrates Raspberry Pi 5 computer vision with an ESP32-S3 sensor node carrying weight, moisture, temperature, and humidity measurements. Its main value is the integration of heterogeneous measurements into a single decision-support workflow rather than the use of any one sensor or AI model in isolation.

The revised methodology defines the data pipeline, calibration procedure, multimodal fusion strategy, experimental design, evaluation metrics, and reporting requirements needed to make the study reproducible. The architecture can support crop-quality grading, moisture classification, storage/transport suitability, and other post-harvest recommendations when crop-specific reference rules are available.

Before submission to a peer-reviewed journal, the remaining experimental fields in Section 4 must be replaced with measurements from the actual prototype and a held-out test set. The final paper should report the dataset size, model metrics, sensor errors, fusion comparison, repeatability, and processing time. These results are essential for converting the present architecture-and-methodology manuscript into a fully validated research paper.

7. Declarations

Funding:

This work did not receive any specific financial support or funding.

Conflict of Interest:

The author declares that there is no conflict of interest related to this work.

Data Availability:

The experimental dataset is not publicly available at this stage. The present work mainly focuses on the proposed system, its design, and implementation framework. Experimental data and detailed performance results will be included in future work after completing the required testing and validation.

Author Contributions:

Bittu Kumar: Designed the concept of the system, developed the methodology, worked on the hardware and software implementation, carried out the analysis, and prepared the manuscript.

Ethics Statement:

Ethical approval was not required for this study because the work focuses on the development of a hardware and software prototype and does not involve human participants, animals, or personal data.

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Appendix A. Minimum Experimental Record for Each Sample

Field	Record
Sample ID	Unique identifier
Crop	Crop class
Batch/lot ID	Physical batch identifier
Image file	Image filename/path
Weight (kg)	Measured value
Moisture (%)	AgriScan value
Reference moisture (%)	Reference value
Temperature (°C)	AgriScan value
Humidity (%)	AgriScan value
Reference quality label	Ground-truth/reference class
AI prediction	Predicted class
Defect label	Reference defect
System grade	A/B/C
Recommendation	Generated recommendation
Processing time (s)	End-to-end time
Notes	Lighting, sample condition, anomalies

8. Pre-Submission Checklist

- Verify all author names, affiliations, email addresses, and corresponding-author details.
- Verify every reference, DOI, volume, issue, and article number against the original publication.
- Insert actual experimental dataset information before making quantitative performance claims.
- Complete sensor calibration and reference-instrument validation before reporting sensor error.
- Add actual confusion matrices, performance metrics, repeatability results, and processing-time measurements if available.
- Check that every figure and table is numbered, cited in the text, and supplied in the format required by the journal.
- Check the current Smart Agricultural Technology Guide for Authors before submission.
- Do not submit illustrative or synthetic numbers as experimental results.